Analysis of Risk-Aware Shortest Path (USEI15)

1 Temporal Complexity Analysis

The implementation of the Bellman-Ford algorithm is divided into three main phases. The total complexity is determined by the dominant step (relaxation).

Phase A: Initialization

Before processing, the algorithm initializes the distance and predecessor maps.

- **Operation:** Set $dist[s] = 0$ and $dist[v] = \infty$ for all other vertices V.
- **Complexity:** $O(V)$
- **Note:** Requires iterating through all vertices to set initial values.

Phase B: Edge Relaxation (Main Loop)

This is the "core" of the algorithm, where edges are processed to find shorter paths.

- **Outer Loop:** Iterates $(V-1)$ times. This is the maximum path length (in edges) possible without forming a cycle.
- **Inner Loop:** In each iteration, the algorithm traverses **all** edges (E) in the graph to check if a shorter path exists (`relax` operation).
- **Operation Cost:** The `relax` method involves basic Map retrieval/update, which is $O(1)$ on average.
- **Complexity:** $O(V \cdot E)$

Phase C: Negative Cycle Detection

After the relaxation phase, one final pass is required to verify if the solution is valid.

- **Operation:** Iterate over all edges (E) one last time. If any edge can still be relaxed ($dist[u] + weight < dist[v]$), a negative cycle exists.
- **Complexity:** $O(E)$

Complexity Conclusion

Combining the three phases:

$$O(V) + O(V \cdot E) + O(E)$$

Since $O(V \cdot E)$ dominates the equation, the Worst-Case Time Complexity is:

$$\mathbf{O(V \cdot E)}$$

Where V is the number of stations and E is the number of connections.